

## Questions

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*Inequalities*

1. For which real values of  $x$  is the following inequality true?

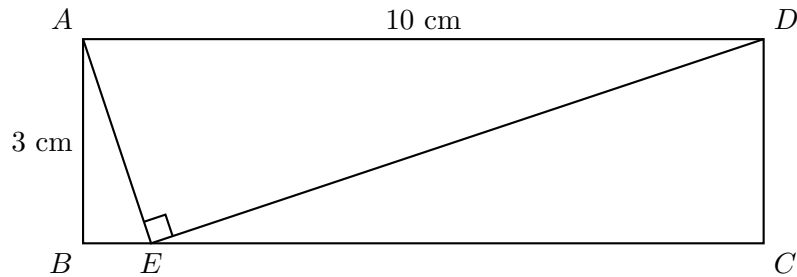
$$x^2 < (x - 3)^2 < (2x + 1)^2$$

- A**  $x < -4$  or  $\frac{2}{3} < x < \frac{3}{2}$   
**B**  $x < -4$  or  $x > \frac{2}{3}$   
**C**  $-4 < x < \frac{2}{3}$   
**D**  $-4 < x < \frac{3}{2}$   
**E**  $\frac{2}{3} < x < \frac{3}{2}$  or  $x > 4$

2.  $ABCD$  is a rectangle with  $AB = 3$  cm and  $AD = 10$  cm.

Point  $E$  lies on  $BC$ , and  $\angle AED = 90^\circ$

What is the value of  $BE \times EC$ ?



- A  $9\text{ cm}^2$
- B  $18\text{ cm}^2$
- C  $41\text{ cm}^2$
- D  $50\text{ cm}^2$
- E  $82\text{ cm}^2$

*Exponentials & Logarithms*

**3.** For  $1 < x < 81$ , the equation

$$\log_2(\log_3 x) + \log_2\left(\log_3 \frac{81}{x}\right) = 1$$

has two solutions. What is the smaller solution?

**A**  $3^{2-\sqrt{3}}$

**B**  $3^{2-\sqrt{2}}$

**C**  $3^{\sqrt{2}}$

**D**  $3^{4-\sqrt{2}}$

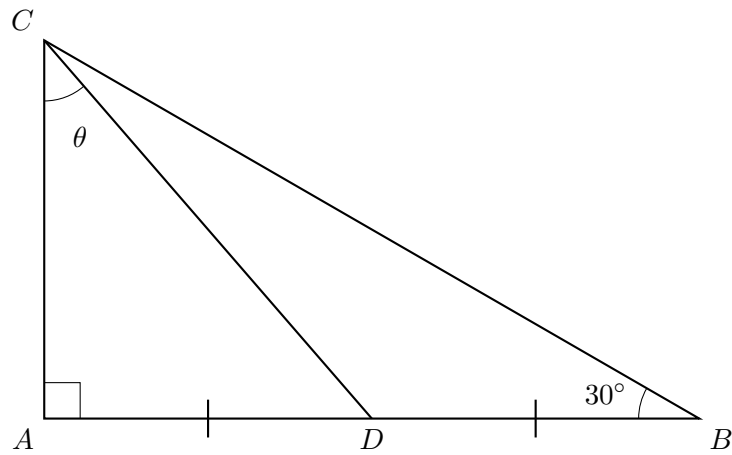
**E**  $3^{2+\sqrt{2}}$

*Trigonometry*

4. Triangle  $ABC$  is right-angled at  $A$ , and angle  $ABC$  is  $30^\circ$

The point  $D$  is the midpoint of  $AB$

If angle  $ACD$  is  $\theta$ , what is  $\cos \theta$ ?



- A  $\frac{1}{\sqrt{7}}$
- B  $\frac{1}{2}$
- C  $\frac{\sqrt{3}}{\sqrt{7}}$
- D  $\frac{2}{\sqrt{7}}$
- E  $\frac{\sqrt{3}}{2}$

*Sequences & Series*

5. An infinite geometric series has positive first term  $a$  and common ratio  $r$ , where  $|r| < 1$ . The sum to infinity of this series is  $S$ .

A second series is formed by taking every third term of the first series, beginning with the first term:

$$T = u_1 + u_4 + u_7 + u_{10} + \dots$$

where  $u_n$  denotes the  $n$ th term of the first series.

What is the maximum possible value of  $\frac{T}{S}$ ?

- A  $\frac{3}{4}$
- B 1
- C  $\frac{7}{6}$
- D  $\frac{4}{3}$
- E  $\frac{3}{2}$
- F 3

*Exponentials & Logarithms*

6. Consider the equation

$$x^{\log_2(x^2)} - 18x^{\log_2 x} + 32 = 0$$

What is the sum of all real solutions to this equation?

- A 1
- B  $\frac{9}{2}$
- C 6
- D  $\frac{25}{4}$
- E  $\frac{13}{2}$
- F  $\frac{27}{4}$
- G  $\frac{31}{4}$
- H 18

*Differentiation*

7. A straight line is tangent to both the curve  $y = x^2$  and the curve  $y = \frac{1}{x}$ , where  $x \neq 0$

At what value of  $x$  does this line cross the  $x$ -axis?

A  $-2$

B  $-\frac{3}{2}$

C  $-1$

D  $-\frac{1}{2}$

E  $-\frac{1}{4}$

*Sequences & Series*

8. Five positive numbers are consecutive terms of a geometric sequence

The middle three have sum 14 and multiply to make 8

What is the sum of all five numbers?

**A** 74

**B** 78

**C** 80

**D** 82

**E** 84

**F** 86

*Probability & Counting*

9. A bag contains 3 red balls and 2 blue balls.

Two balls are drawn without replacement.

If they are the same colour, both are returned to the bag. If they are different colours, both are left out.

A ball is then drawn from the bag.

What is the probability that this final ball is red?

- A  $\frac{3}{5}$
- B  $\frac{19}{30}$
- C  $\frac{16}{25}$
- D  $\frac{2}{3}$
- E  $\frac{7}{10}$

*Algebra*

**10.** Real numbers  $x$  and  $y$  satisfy

$$x + y = xy$$

and

$$(x - 1)^2 + (y - 1)^2 = 7$$

What are the possible values of  $xy$ ?

**A**    $-5$  and  $1$

**B**    $-3$  and  $3$

**C**    $-1$  and  $3$

**D**    $-1$  and  $5$

**E**    $1$  and  $5$

*Probability & Counting*

11. Find the coefficient of  $x^3y^2$  in the expansion of

$$(1 - 2x + y)^8$$

**A**     $-17920$

**B**     $-13440$

**C**     $-4480$

**D**     $-560$

**E**     $560$

**F**     $4480$

**G**     $17920$

**12.** The function  $f$  is defined for all real  $x$  by

$$f(x) = \frac{x^2 + 4x + 1}{x^2 + 1}$$

Which of the following gives all possible values of  $f(x)$  as  $x$  varies over the real numbers?

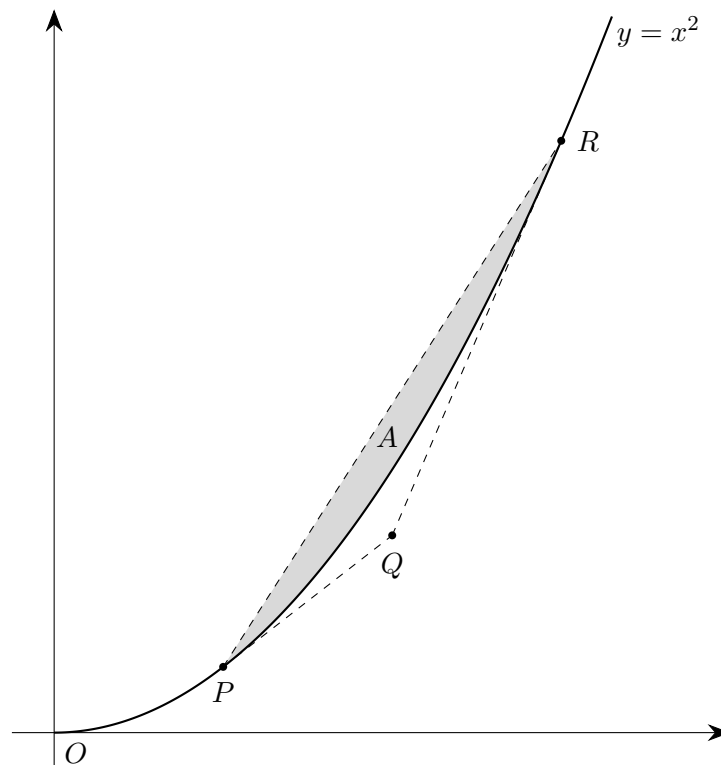
- A**  $-1 \leq f(x) \leq 3$
- B**  $f(x) \leq -1$  or  $f(x) \geq 3$
- C**  $1 \leq f(x) \leq 3$
- D**  $-3 \leq f(x) \leq 1$
- E**  $-1 \leq f(x) \leq 1$
- F**  $f(x)$  can take all real values

## Coordinate Geometry

13. The points  $P$  and  $R$  lie on the curve  $y = x^2$  and have  $x$ -coordinates 1 and 3, respectively. The tangents to the curve at  $P$  and  $R$  meet at  $Q$ .

The area enclosed by the curve and the line segment  $PR$  is  $A$ . The area of triangle  $PQR$  is  $B$ .

What is the value of  $\frac{A}{B}$ ?



- A  $\frac{1}{3}$
- B  $\frac{1}{2}$
- C  $\frac{2}{3}$
- D  $\frac{3}{4}$
- E  $\frac{3}{2}$

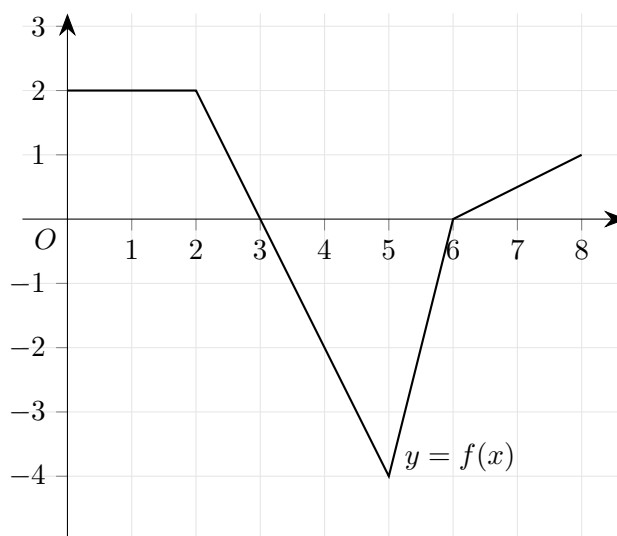
## Integration

14. The diagram shows the graph of  $y = f(x)$  for  $0 \leq x \leq 8$ . The graph is made up of straight line segments joining the points  $(0, 2)$ ,  $(2, 2)$ ,  $(5, -4)$ ,  $(6, 0)$  and  $(8, 1)$ .

The function  $g$  is defined by

$$g(x) = \int_0^x f(t) \, dt \quad \text{for } 0 \leq x \leq 8$$

How many values of  $x$  with  $0 \leq x \leq 8$  satisfy  $g(x) = 0$ ?



- A 0
- B 1
- C 2
- D 3
- E 4
- F 5

*Number*

**15.** Let  $k$  be a positive integer, and let  $N = 2^4 \times 3^k$

The product of all the positive factors of  $N$  is the fourth power of an integer.

What is the smallest possible value of  $k$ ?

**A** 1

**B** 2

**C** 3

**D** 5

**E** 7

**F** 8

*Polynomials*

- 16.** A quadratic polynomial  $f$  has roots 1 and 3.

The equation  $f(f(x)) = 0$  has exactly three distinct real solutions.

What is the value of  $f(0)$ ?

- A**    $-9$
- B**    $-3$
- C**    $-1$
- D**    $3$
- E**    $9$

*Trigonometry*

17. A triangle has an angle of  $120^\circ$

The perimeter of the triangle is 15, and its area is  $\frac{15\sqrt{3}}{4}$

What is the length of the longest side of the triangle?

**A** 5

**B**  $\frac{11}{2}$

**C** 6

**D**  $\frac{13}{2}$

**E** 7

**F**  $\frac{15}{2}$

**G** 8

*Coordinate Geometry*

18. The point  $A$  has coordinates  $(-7, -5)$  and the point  $B$  has coordinates  $(23, 45)$ .

How many points on the straight line segment  $AB$  have both coordinates equal to integers, not counting  $A$  and  $B$  themselves?

- A 4
- B 5
- C 9
- D 10
- E 11
- F 29
- G 49

*Algebra*

**19.** The positive real numbers  $x$  and  $y$  satisfy the equations

$$x\sqrt{y} + y\sqrt{x} = 8$$

and

$$x\sqrt{x} + y\sqrt{y} = 40$$

What is the value of  $x + y$ ?

- A** 6
- B** 8
- C** 10
- D** 12
- E** 14
- F** 16

**20.** Let  $f$  be a polynomial of degree at most 3 with real coefficients.

For  $t > 0$ , define

$$D(t) = \int_0^t f(x) \, dx - \int_{-t}^0 f(x) \, dx$$

Which of the following statements are true?

- I** If  $f(x) = f(-x)$  for every real  $x$ , then  $D(1) = 0$
  - II** If  $D(1) = 0$ , then  $f(x) = f(-x)$  for every real  $x$
  - III** If  $D(1) = D(2) = 0$ , then  $f(x) = f(-x)$  for every real  $x$
- 
- A** None of them
  - B** I only
  - C** II only
  - D** III only
  - E** I and II only
  - F** I and III only
  - G** II and III only
  - H** I, II and III